Lesson 8: Motors and Left Hand Rule



Teacher: Dan (1 Practical)

GCSE Physics: Electricity and Magnetism

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Luminous Blueprint

What is the motor effect?

Success Cue: Field interaction creates force

Answer: A current-carrying wire in a magnetic field experiences a mechanical force.

State Fleming's Left Hand Rule for thumb, first finger, second finger.

Success Cue: F-B-I / Motion-Field-Current

Answer: Thumb = Motion/Force, First finger = Field (N → S), Second finger = Current (+ → −)

What formula calculates the force on a conductor in a magnetic field?

Success Cue: Force = Flux density × Current × Length

Answer: F = B × I × L

What is the unit of magnetic flux density (B)?

Success Cue: Symbol T

Answer: Tesla (T)

What is the vital function of the split-ring commutator in a DC motor?

Success Cue: Prevents motor stalling

Answer: Reverses current direction every half-turn to keep coil rotating in same direction.

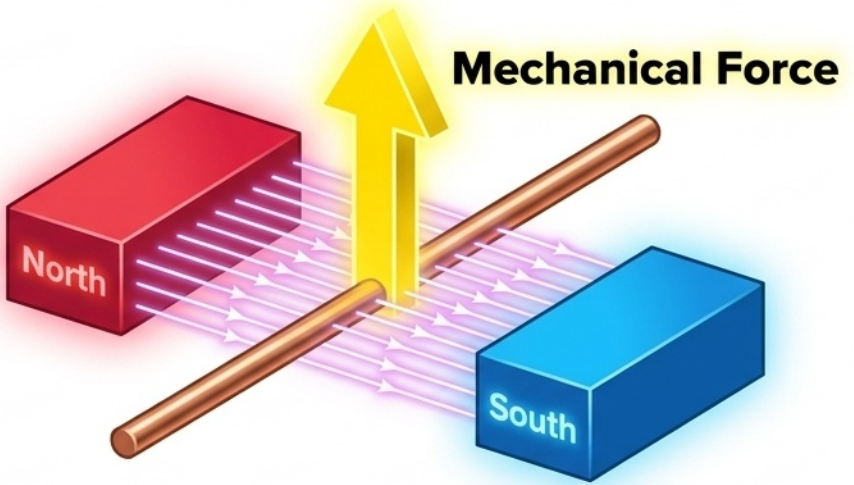
Name two ways to increase the turning effect in a DC motor.

Success Cue: Think variables

Answer: Larger current, stronger magnetic field, or more turns in the coil.

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The Core Phenomenon: The Motor Effect



- **Current + Magnetism: A current-carrying conductor placed in a magnetic field experiences a force.**
- **Field Interaction: The wire's own magnetic field interacts with the permanent magnets.**
- **The Result: This interaction produces a distinct mechanical push on the wire.**

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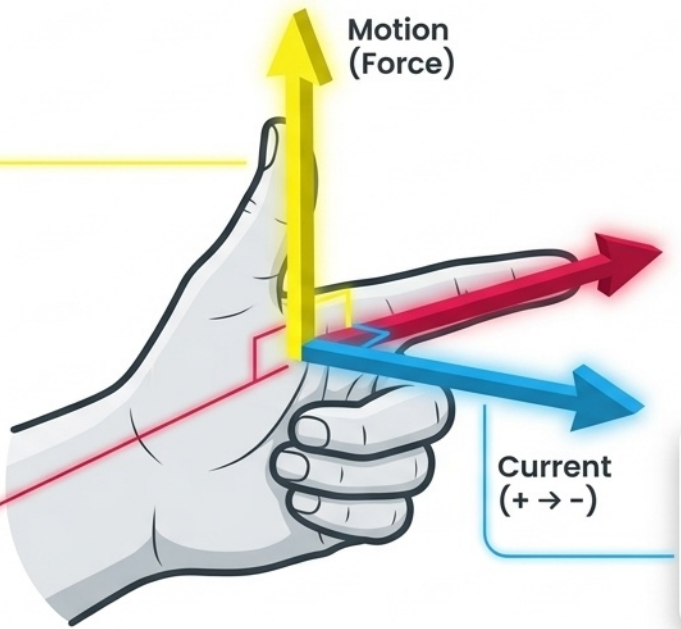
Predicting Force: Fleming's Left Hand Rule

ThuMb

Represents the direction of Motion (or Force).

First Finger

Represents the Magnetic Field (North to South).



seCond Finger

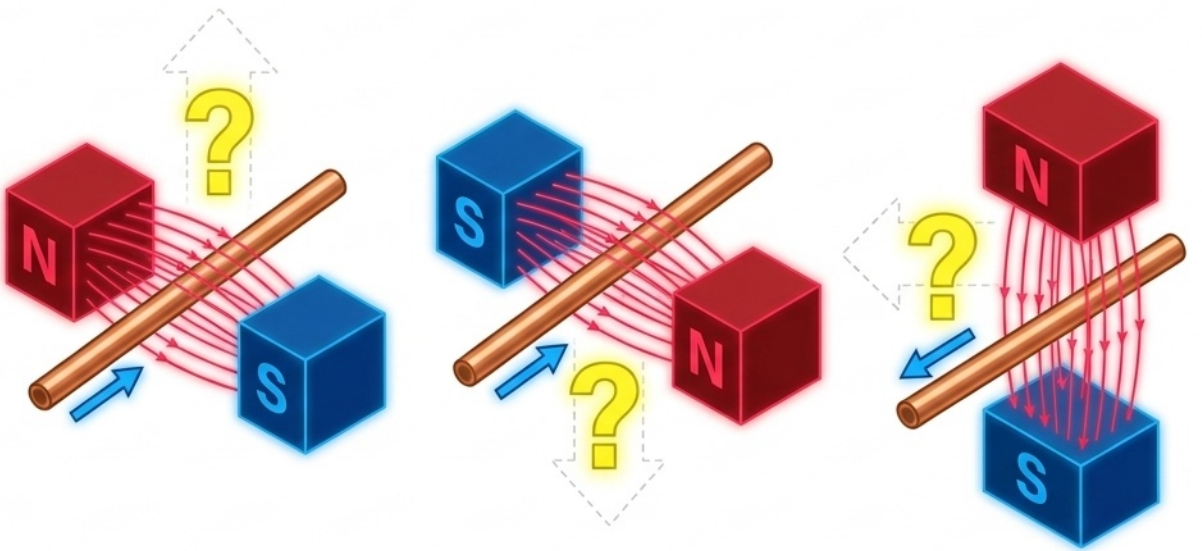
Represents conventional Current (+ to -).

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Knowledge Checkpoint: The Finger Gym

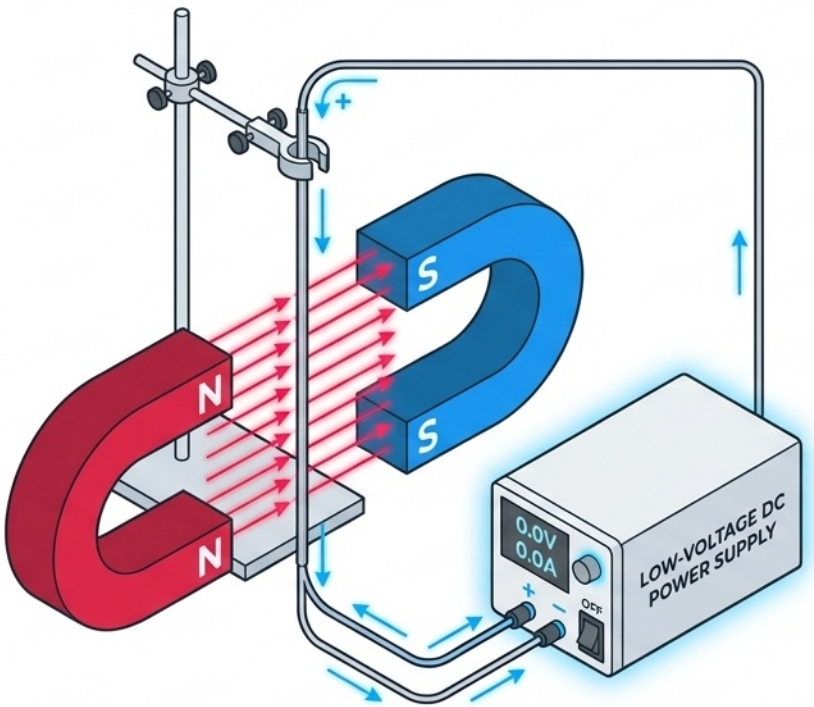
Pair and Share: Use your left hand to predict the direction of force for each wire.

- Align your First Finger (N→S).
- Align your Second Finger (Current).
- Where does your Thumb point?



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Mandatory Practical 1: Setup & Method



Equipment

Rigid wire, strong permanent magnets, low-voltage power supply.

Safety

Use low voltage to prevent wire overheating. Do not leave circuit on continuously.

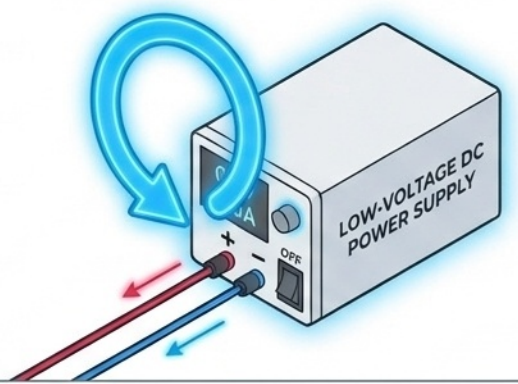
Step 1 Action

Suspend the wire between the poles. Switch on the power supply to pass a current.

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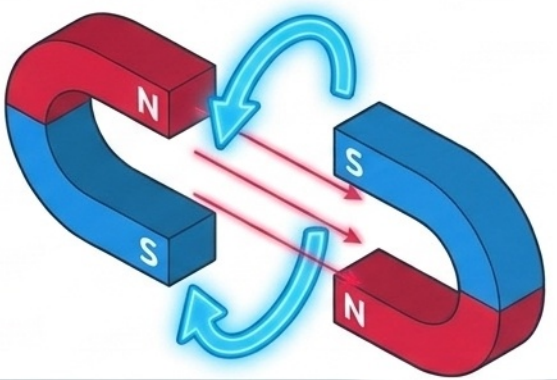
Mandatory Practical 1: Observation & Reversal

Reversing the Current



Action: Swap the positive and negative connections.
Observation: The wire deflects in the exact opposite direction.

Reversing the Magnetic Poles

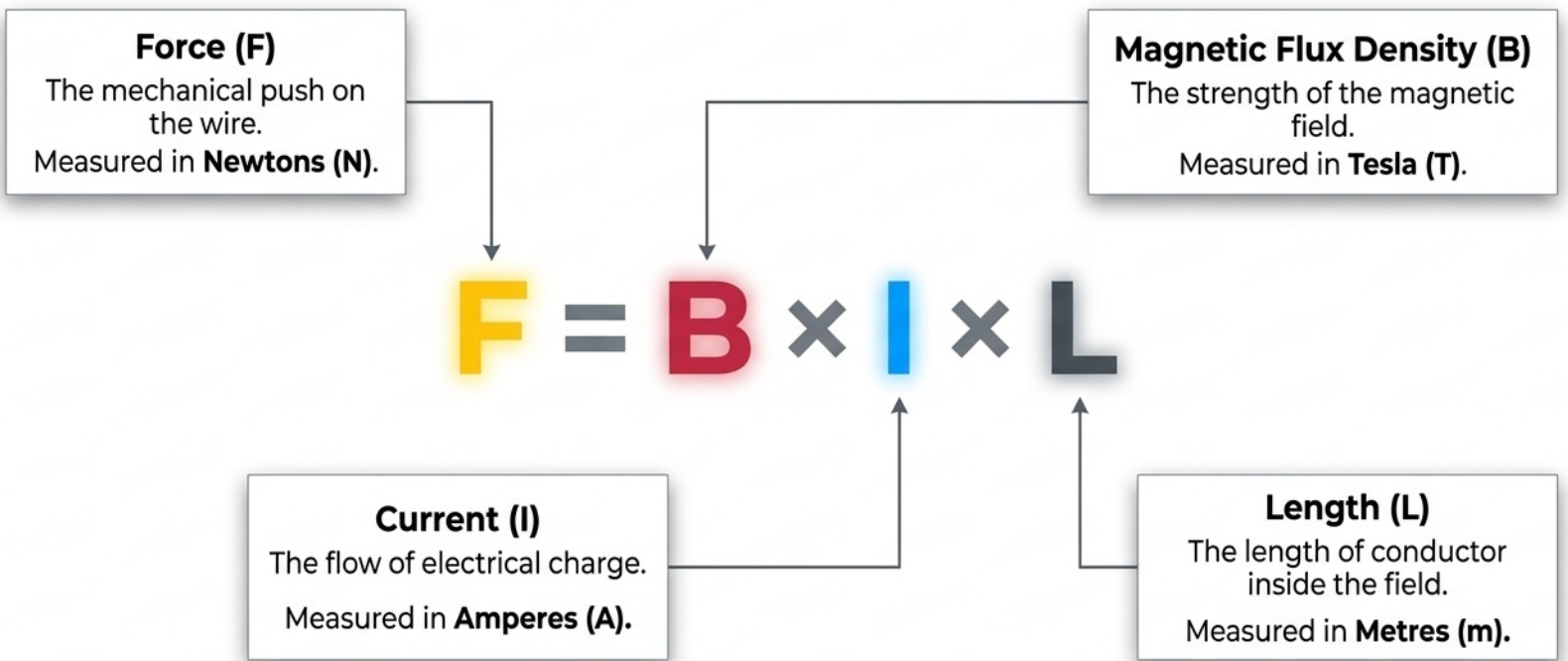


Action: Flip the magnets so South replaces North.
Observation: The wire deflects in the opposite direction again.

Task: Use Fleming's Left Hand Rule to verify both of these observations!

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Quantifying the Force: The Equation



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Knowledge Checkpoint: Calculation Race

Teacher Model

Prompt: A 0.5m wire carries 2A in a 0.3T magnetic field. Calculate Force.

Formula: $F = B \times I \times L$

Substitute: $F = 0.3 \times 2 \times 0.5$

Result: $F = 0.3$ N

Your Turn



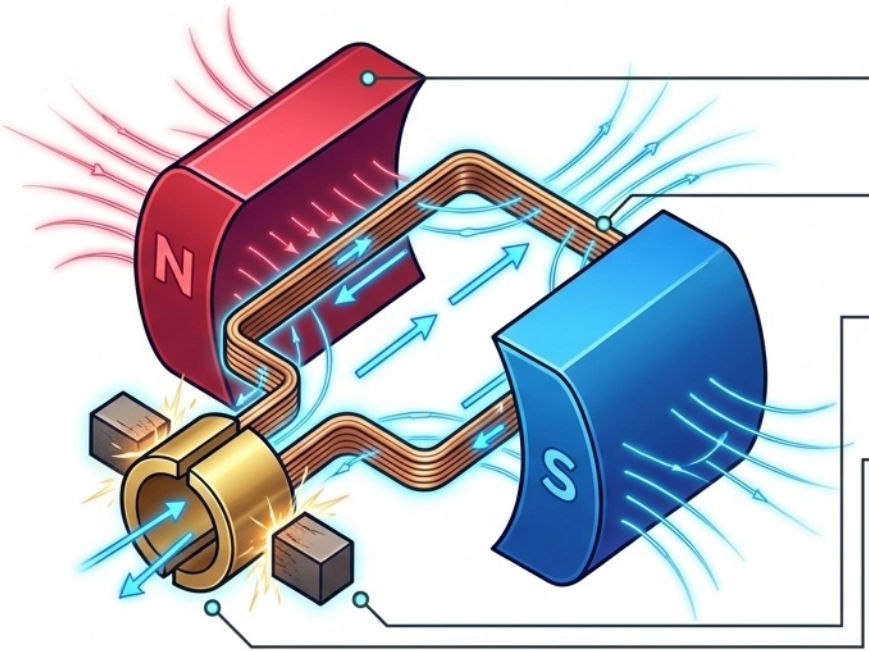
Prompt: A 2.0m wire carries 5A in a 0.4T magnetic field. Calculate Force.

Steps: Formula → Substitute → Calculate.

Answer: $F = 0.4 \times 5 \times 2.0 = 4.0$ N

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Engineering Application: The DC Electric Motor



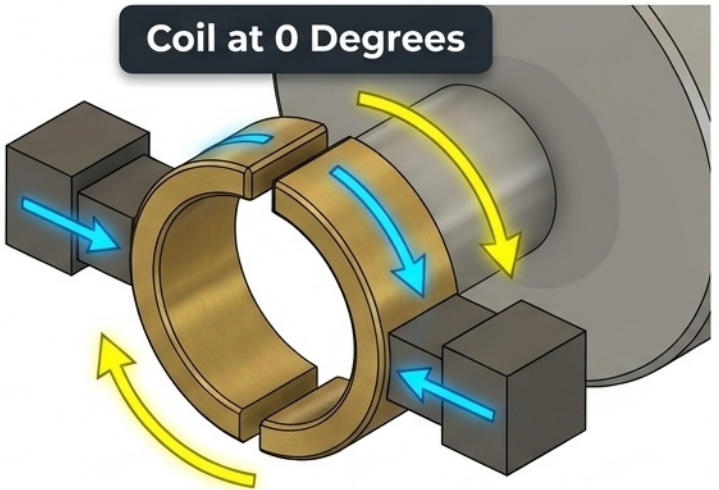
1. **Permanent Magnets:**
   Provide the external magnetic field.
2. **Rectangular Coil:**
   A loop of wire that carries the current.
3. **Carbon Brushes:**
   Provide sliding electrical contact to the rotating coil.
4. **Split-ring Commutator:**
   (Detailed on next slide)

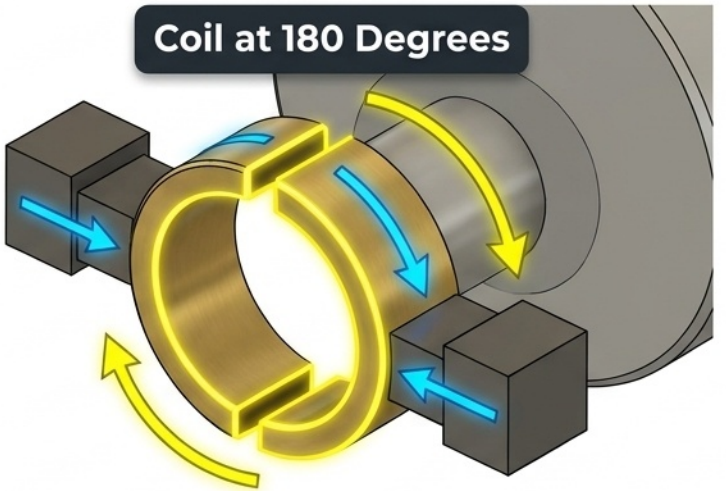
Core Principle

Forces on opposite sides of the coil act in opposite directions, producing continuous rotation.

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The Clever Trick: The Split-Ring Commutator





The Problem: Without it, the coil would just flip vertically and stall.

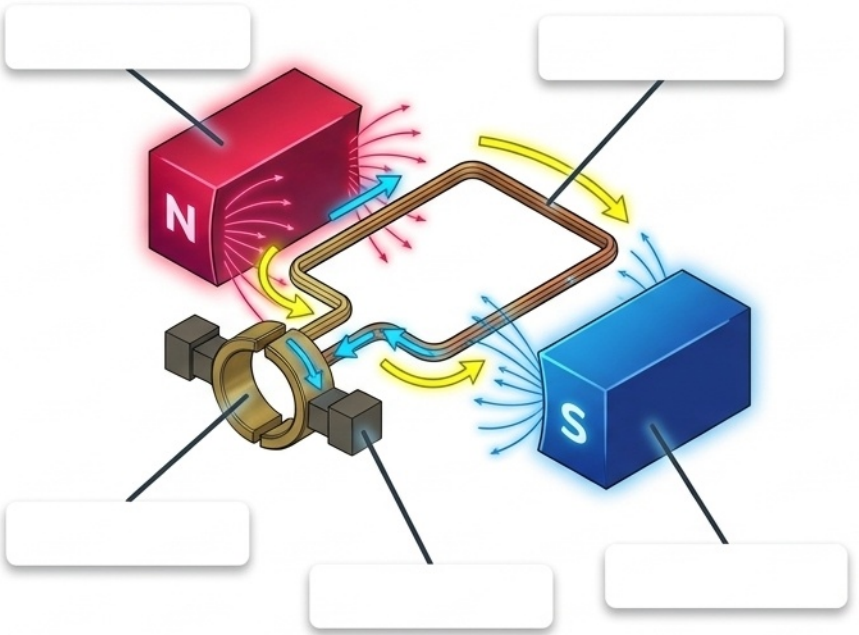
The Solution: A **split ring** that swaps electrical contacts every half-turn.

The Result: It reverses the current direction in the coil every half-turn, keeping the motor spinning in the same direction.

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Synthesis Checkpoint: Match & Label

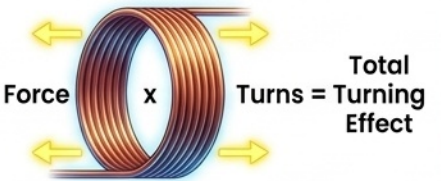
Instructions: Pair-and-share! Race to accurately label the parts and match them to their function.



Word Bank	Function Bank
Commutator	Provides external field.
Brushes	Carries the current.
Coil	Maintains electrical contact.
Permanent Magnets	Reverses current every half-turn.

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Optimising Performance: Upgrading the Motor

The Variable	The Action	The Physics Result
Current	Increase it (use a higher voltage).	Creates a stronger electromagnetic field around the wire. 
Magnetic Field	Use stronger magnets.	Increases the flux density (B). 
Coil	Add more turns of wire.	Multiplies the force; each turn adds to the total turning effect. 

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Lesson Plenary: 3-2-1 Summary

1

Formula for Magnetic Force

F = B × I × L (Force = Flux Density × Current × Length).

2

Fingers used for Fields (LHR)

- First Finger = Magnetic Field (N→S).
- Second Finger = Current (+→-).

3

Key Components of a DC Motor

- Coil, Split-ring commutator, Carbon brushes.

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